



Linux Interview Question

I. What is the boot process in Linux?

BIOS

Basic Input/Output System
executes MBR

MBR

Master Boot Record
executes GRUB

GRUB

Grand Unified Bootloader
executes Kernel

thegeekstuff.com

Kernel

Kernel
executes /sbin/init

Init

Init
executes runlevel programs

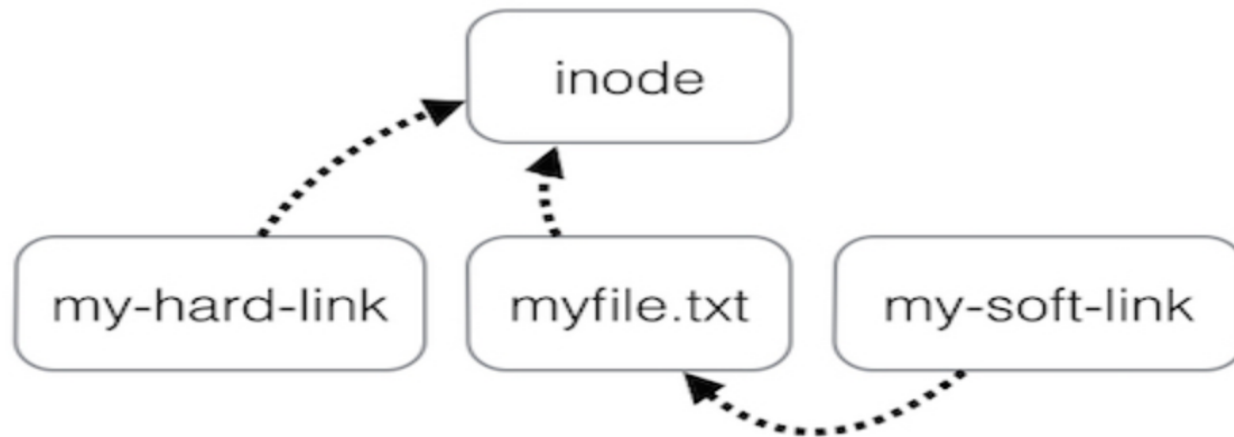
Runlevel

Runlevel programs are
executed from /etc/rc.d/rc*.d/

2. How to create zero size file Linux?

- Using touch command we can create zero size file

3. What is soft link and hard link?
How to create? What is the
difference between these two?





Using ln command we will create link

```
$ ln myfile.txt my-hard-link
```

```
$ ln -s myfile.txt my-soft-link
```

4. What is first line written in shell script?
What is the meaning of that? If I didn't write that line what will happen? Then how to run the script?

- `#!/bin/bash` – Execute the file using the Bash shell.
- `#!` Called shebang
- The `#!` syntax used in scripts to indicate an interpreter for execution under UNIX or Linux operating systems
- If not written in file we need to execute like below

```
/bin$ bash /home/madhu/test.sh
```

5. How to run a shell script in background?

- Execute a command in the background using **&**
`$ ./my-shell-script.sh &`
- 2. Execute a command in the background using **nohup**
- After you execute a command (or shell script) in the background using **&**, if you logout from the session, the command will get killed. To avoid that, you should use **nohup** as shown below.
- `$ nohup ./my-shell-script.sh &`

6. What is cron tab? Explain it? How to configure the schedule a job?

- crontab uses a daemon, crond, which runs constantly in the background and scheduled jobs need to be executed.
- MIN HOUR DOM MOY DOW
* * * * *
- 0-59 0-23 1-31 1-12 0-6
- To check crontablist “ crontab -l”
- To edit crontab “crontab -e”

7.How to allow the ports in Linux?

- `iptables -A INPUT -p tcp --dport 12375 -j ACCEPT`
- `iptables -A OUTPUT -p tcp --dport 12375 -j ACCEPT`
- you can also add them to your iptables-config file then restart iptables

`/etc/init.d/iptables restart`

8. How to trouble shoot the remote server having some issue?

- First will do ssh if it's connected
- Uptime
- Top
- Free
- df
- du

9. What is ping? Telnet? Curl? Wget?

- **Ping** commonly used to check whether your **connection to the server** is healthy or not.
- **Telnet** is a user command and an underlying TCP/IP protocol for accessing remote computers.
- **wget** is a tool to download files from servers.
- **curl** is a tool that let's you exchange requests/responses with a server

10. How to check the services in Linux machine?

- `chkconfig --list`

11. How to kill the process in Linux?

- First check the whichh process you want to kill using ps command. Then will get pid
- Kill -9 <pid>

12. What is nice and renice?

- You can ask the kernel to put a particular process of yours, on higher priority than others using **nice** command
- Process priority values range from **-20 to 19** **nice value**.

nice -10 <command name>

- In order to change the priority of an already running process you can use "renice" command.
- **renice 13 -p <PID>**
- -20 (most favorable to the process) to 19 (least favorable to the process)

13. What is inode value?

- An inode is an entry in inode table, containing information (the metadata) about a regular file and directory. An inode is a data structure on a traditional Unix-style file system such as ext3 or ext4

- `$ ls -il myfile.txt`

```
1150561 -rw-r--r-- 1 root root 0 Mar 10
01:06 myfile.txt
```

I 4. How to check the CPU utilization?

- Using top command

15. Difference between Top/HTop?

- In 'htop' you can scroll the list vertically and horizontally to see all processes and complete command lines.
- In 'top' you are subject to a delay for each unassigned key you press (especially annoying when multi-key escape sequences are triggered by accident).
- 'htop' starts faster ('top' seems to collect data for a while before displaying anything).
- In 'htop' you don't need to type the process number to kill a process, in 'top' you do.
- In 'htop' you don't need to type the process number or the priority value to renice a process, in 'top' you do.
- 'htop' supports mouse operation, 'top' doesn't
- 'top' is older, hence, more used and tested.

16. What is mount? How to create mount?

- Mounting a filesystem means making the particular filesystem accessible at a certain point in the Linux directory tree.
- **\$ mount**
- When you type this at a command prompt, this command will display all the mounted devices, the filesystem type it is mounted as, and the mount point.
- `mount [OPTION...] DEVICE_NAME DIRECTORY`
- <https://linuxize.com/post/how-to-mount-and-unmount-file-systems-in-linux/>

17. How to trouble shoot live logs?

- Using “tail -f filename”

18. What is sed command?

- SED command in UNIX is stands for stream editor and it can perform lot's of function on file like, searching, find and replace, insertion or deletion.
- **`$sed 's/unix/linux/' file.txt`**

19. What is AWK command?

- Awk is a scripting language used for manipulating data and generating reports. The awk command programming language requires no compiling, and allows the user to use variables, numeric functions, string functions, and logical operators.
- Awk is abbreviated from the names of the developers – Aho, Weinberger, and Kernighan.
- <https://www.geeksforgeeks.org/awk-command-unixlinux-examples/>

20. What is grep and egrep?

- grep command we will use to search particular content.

grep madhu filename.txt

- Egrep we will use you want grep multiple word at a time
- egrep 'madhu|sudhan' filename.txt

21. How to list out the only directories in a Linux?

- `$ ls -l | grep '^d'`

22. How to check the process in Linux?

- Using ps command

23. How to get the java thread dump?

- Using jstack command

24. How to check the running ports?

- Using **netstat -tulpn | grep LISTEN**

25. How to declare a variable in a shell script?

```
days=5
```

```
name="madhu"
```

26. What is \$?, \$#, \$*, \$@ ?

- \$# Stores the number of command-line arguments that were passed to the shell program.
- \$? Stores the exit value of the last command that was executed. (0 or 1)
- \$0 Stores the first word of the entered command (the name of the shell program).
- \$* Stores all the arguments that were entered on the command line (\$1 \$2 ...).
- "\$@" Stores all the arguments that were entered on the command line, individually quoted ("\$1" "\$2" ...).



`./test.sh one two "three four"`

- Using `$*`:

one

two

three

four

- Using `"$@"`:

one

two

three four

27. How to read a command line input in shell script?

- with `$<argumentnumber>`
- `$1` or `$2` or `$3`

28. What is umask?

- Return, or set, the value of the system's file mode creation mask.
- In Linux, the default permissions value is **666** for a regular file, and **777** for a directory. When creating a new file or directory, the kernel takes this default value, "subtracts" the umask value, and gives the new files the resulting permissions. ($666 - 022 = 644$)
- /etc/profile.d
- <https://www.computerhope.com/unix/uumask.htm>

29. How to change file permission in Linux?

- using `chmod` command.
- the **u**ser can **r**ead, **w**rite, and **x**ecute it;
- members of your **g**roup can **r**ead and **x**ecute it; and
- **o**thers may only **r**ead it.

`chmod u=rwx,g=rx,o=r myfile`

`chmod 754 myfile`

- **4** stands for "read",
- **2** stands for "write",
- **1** stands for "execute", and
- **0** stands for "no permission."

30. How to connect remote servers without password? How to archive this?

- using ssh-keygen
- To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool. Like this:
- `ssh-copy-id -i ~/.ssh/tatu-key-ecdsa user@host`

31. How to open file in read only mode in VI editor?

- **vi -R filename**

32. What is the purpose of export command?

- **export** - Set export attribute for shell variables.

Example:

To Set JAVA_HOME:

```
$ export JAVA_HOME=/usr/local/jdk
```

33. How to send error logs and stdout logs in to files?

- There are two main output streams in Linux (and other OSs), standard output (stdout) and standard error (stderr). Error messages, like the ones you show, are printed to standard error. The classic redirection operator (command > file) only redirects standard output, so standard error is still shown on the terminal. To redirect stderr as well, you have a few choices:

- Redirect stdout to one file and stderr to another file:

command > out 2>error

- Redirect stderr to stdout (&1), and then redirect stdout to a file:

command >out 2>&1

- Redirect both to a file:

command &> out



34. What is nohup command?

- to run a script in background

35. What is netstat command in Linux?

- netstat (network statistics) is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc.

netstat -a | more : To show both listening and non-listening sockets.

netstat -at : To list all tcp ports.

netstat -au : To list all udp ports.

netstat -l : To list only the listening ports.

netstat -lt : To list only the listening tcp ports.

netstat -lu : To list only the listening udp ports.

netstat -lx : To list only the listening UNIX ports.

netstat -s : To list the statistics for all ports.

netstat -an | grep ':80' : To get the process which is using the given port.

36. How to run a script at boot level?

- In the file you put in /etc/init.d/ you have to set it executable with:

```
chmod +x /etc/init.d/start_my_app
```

- If this does not run you have to create a symlink to /etc/rc.d/

```
ln -s /etc/init.d/start_my_app /etc/rc.d/
```